

ANSWER SHEET

1.a. Data type of all columns in the "customers" table.

Ans

```
SELECT
column_name,
data_type
FROM
`case-study-463713.Target.INFORMATION_SCHEMA.COLUMNS`
WHERE
table_name = 'customers';
```

Row	column_name	data_type
1	customer_id	STRING
2	customer_unique_id	STRING
3	customer_zip_code_prefix	INT64
4	customer_city	STRING
5	customer_state	STRING

Insights:

- There are a total of 5 columns consisting of 4 string datatypes and one integer datatypes

Remarks:

- N/A

1.b. Get the time range between which the orders were placed.

Ans

```
SELECT
MIN(order_purchase_timestamp) as first_order_date,
MAX(order_purchase_timestamp) as last_order_date
FROM
`case-study-463713`.`Target`.`orders`;
```

Row	first_order_date ▼	last_order_date ▼
1	2016-09-04 21:15:19 UTC	2018-10-17 17:30:18 UTC

Insights:

- Orders are placed between the year 2016 September and 2018 October

Remarks:

- N/A

1.c. Count the Cities & States of customers who ordered during the given period.

Ans

```
select
  customer_city,
  customer_state,
  count(distinct c.customer_id) as no_orders
from `Target.customers` as c
inner join `Target.orders` as o on c.customer_id=o.customer_id
group by customer_city, customer_state
order by customer_state,customer_city,no_orders desc
```

Row	customer_city ▼	customer_state ▼	no_orders ▼
1	brasileia	AC	1
2	cruzeiro do sul	AC	3
3	epitaciolandia	AC	1
4	manoel urbano	AC	1
5	porto acre	AC	1
6	rio branco	AC	70
7	senador guiomard	AC	2
8	xapuri	AC	2
9	agua branca	AL	1
10	anadia	AL	2
11	arapiraca	AL	29

Insights:

- The city with highest number of orders is “Maceio” with 247 orders

Remarks:

- N/A

2.a. Is there a growing trend in the no. of orders placed over the past years?

Ans

```
SELECT
    extract (year from order_purchase_timestamp) AS order_year,
    COUNT(order_id) AS order_count
FROM
    `case-study-463713`.`Target`.`orders`
GROUP BY
    order_year
ORDER BY
    order_year ASC;
```

Row	order_year	order_count
1	2016	329
2	2017	45101
3	2018	54011

Insights:

- yes, there seems to be good growing trend from 2016 to 2017, and it is still growing in 2018 as well.
- The business seems to be struggling at the beginning of the year but from the later years onwards it started improving

Remarks:

- Try to understand the changes that were made over the year 2016 and 2017 note the key attributes which changed the order sales.

2.b Can we see some kind of monthly seasonality in terms of the no. of orders being placed?

Ans

```
SELECT
  extract( month from order_purchase_timestamp) AS month,
  COUNT(*) AS num_orders
FROM
  `case-study-463713`.`Target`.`orders`
GROUP BY
  month
ORDER BY
  month;
```

Row	month	num_orders
1	1	8069
2	2	8508
3	3	9893
4	4	9343
5	5	10573
6	6	9412
7	7	10318
8	8	10843
9	9	4305
10	10	4959
11	11	7544

Insights:

- It is noticed that there is monthly seasonality trend in the sales.
- The highest sales made are in the month's autumn and winter (6-8 month).
- The least sales made are during the months of Spring (9-11).
- The sales seem to be balanced during the season of summer.

Remarks:

- Since the sales are dropping significantly high on month of September and October and the sales are high in winter season, we can try to sell more products on the winter season and clear our warehouse stocks by giving some offers such as buy one get one free.
- Reduce the number of employees in warehouse or store during the Spring (9-11 months) season and increase the number of employees during the winter season (6-8 months) to achieve more profits.

2.c. During what time of the day, do the Brazilian customers mostly place their orders? (Dawn, Morning, Afternoon or Night)

- 0-6 hrs : Dawn
- 7-12 hrs : Mornings
- 13-18 hrs : Afternoon
- 19-23 hrs : Night

Ans

```
SELECT
CASE
WHEN EXTRACT(HOUR FROM order_purchase_timestamp) BETWEEN 0 AND 6 THEN 'Dawn'
WHEN EXTRACT(HOUR FROM order_purchase_timestamp) BETWEEN 7 AND 12 THEN 'Morning'
WHEN EXTRACT(HOUR FROM order_purchase_timestamp) BETWEEN 13 AND 18 THEN 'Afternoon'
ELSE 'Night'
END
AS time_of_day,
COUNT(customer_id) AS order_count
FROM
`case-study-463713`.`Target`.`orders`
GROUP BY
time_of_day
ORDER BY
order_count DESC;
```

Row	time_of_day ▼	order_count ▼
1	Afternoon	38135
2	Night	28331
3	Morning	27733
4	Dawn	5242

Insights:

- It's been observed that the highest number of orders are placed during the afternoon.
- There are good number of orders placed during morning and night as well and the least number of orders are placed during dawn

Remarks:

- most customers prefer shopping or completing transactions post-lunch hours possibly when they're free from work or household tasks.
- morning activity is also high which indicate most of the people start they're by browsing after waking up or after breakfast.
- Least activity is at dawn which could be mostly because people might be sleeping but it is noticeable there are small groups of people still browsing during dawn mostly after works or possibly people preferring to spend their time during late nights.

3.a Get the month-on-month no. of orders placed in each state.

Ans

```

SELECT
extract(month from orders.order_purchase_timestamp) AS order_month,
customers.customer_state,
COUNT(*) AS num_orders
FROM
`case-study-463713`.`Target`.`orders` AS orders
LEFT JOIN
`case-study-463713`.`Target`.`customers` AS customers
ON
orders.customer_id = customers.customer_id
GROUP BY
order_month,
customers.customer_state
ORDER BY

```

```
order_month,  
customers.customer_state;
```

Row	order_month ▾	customer_state ▾	num_...
1	1	AC	8
2	1	AL	39
3	1	AM	12
4	1	AP	11
5	1	BA	264
6	1	CE	99
7	1	DF	151
8	1	ES	159
9	1	GO	164
10	1	MA	66
11	1	MG	971

Insights:

- It is noted that the state SP (Sao Paulo) is generating the most sales in every month mostly because of the highest population in Brazil is in SP.
- It is also observed all over the states sales are high between the months of January to august (1-8) and sales dropping highly during spring (9-10).
- Least sales produced state is RR(Roraima)

Remarks:

- It can be noticed that the state with highest population is making more sales (SP) and the state with least population is making least sales (RR).
- Thus, we can focus advertising more on states with higher population to gain more attraction from customers.
- Since the month of Spring (9 and 10) is having less sales, we can reduce the advertisement during these months and focus more on advertisements during pre-spring which is having highest sales of the month (7 and 8).

3.b How are the customers distributed across all the states?

Ans

```
SELECT
    customers.customer_state,
    COUNT(DISTINCT customers.customer_id) AS num_customers
FROM
    `case-study-463713`.`Target`.`customers` AS customers
GROUP BY
    1;
```

Row	customer_state ▼	num_customers ▼
1	AC	81
2	AL	413
3	AM	148
4	AP	68
5	BA	3380
6	CE	1336
7	DF	2140
8	ES	2033
9	GO	2020
10	MA	747
11	MG	11635

Insights:

- Highest number of customers are observed in state SP consisting of 41K customers followed by RJ and MG consisting of 12.85k and 11.64K respectively.
- The least number of customers is observed to be in RR consisting of only 46 customers between the years 2016 and 2018

Remarks:

- It can be noted that the state with higher population is having more customers to the company and thus the company Target is preferred to focus their business mostly on highly populated areas.

4.a Get the % increase in the cost of orders from year 2017 to 2018 (include months between Jan to Aug only).

You can use the "payment value" column in the payments table to get the cost of orders.

Ans

```
WITH ORDER_2017 AS (
    select extract(year from order_purchase_timestamp ) as year,
    round(sum(payment_value),2) as Totalcost_2017 from `Target.orders` as o
    inner join `Target.payments` as p on o.order_id=p.order_id
    where extract(month from order_purchase_timestamp ) between 1 and 8
    and extract(year from order_purchase_timestamp )= 2017
    group by extract(year from order_purchase_timestamp )
),
ORDER_2018 AS (
    select extract(year from order_purchase_timestamp ) as year,
    round(sum(payment_value),2) as Totalcost_2018 from `Target.orders` as o
    inner join `Target.payments` as p on o.order_id=p.order_id
    where extract(month from order_purchase_timestamp ) between 1 and 8
    and extract(year from order_purchase_timestamp )= 2018
    group by extract(year from order_purchase_timestamp )
)

SELECT Totalcost_2017,Totalcost_2018,round(((Totalcost_2018-
Totalcost_2017)/Totalcost_2017)*100,2) as percentage_increase FROM
ORDER_2017,ORDER_2018
```

Row	Totalcost_2017	Totalcost_2018	percentage_incre...
1	3669022.12	8694733.84	136.98

Insights:

- The percentage cost from 2017 to 2018 and only between the months 1-8) is 136.98.
- This shows the Profits made by the Target firm is usually between the months 1-8

Remarks:

- The company should try to achieve more sales during this month range as it is observed to be a seasonality trend
- Its recommended at the end of the month like on month 8 the company should try to clear their stock

4.b Calculate the Total & Average value of order price for each state.

Ans

```
SELECT
    customers.customer_state,
    round(SUM(order_items.price),2) AS total_order_price,
    round(AVG(order_items.price),2) AS average_order_price
FROM
    `case-study-463713`.`Target`.`orders` AS orders
INNER JOIN
    `case-study-463713`.`Target`.`customers` AS customers
ON
    orders.customer_id = customers.customer_id
INNER JOIN
    `case-study-463713`.`Target`.`order_items` AS order_items
ON
    orders.order_id = order_items.order_id
GROUP BY
    1
ORDER BY
    customers.customer_state;
```

Row	customer_state	total_order_price	average_order_price
1	AC	15982.95	173.73
2	AL	80314.81	180.89
3	AM	22356.84	135.5
4	AP	13474.3	164.32
5	BA	511349.99	134.6
6	CE	227254.71	153.76
7	DF	302603.94	125.77
8	ES	275037.31	121.91
9	GO	294591.95	126.27
10	MA	119648.22	145.2
11	MG	1585308.03	120.75

Insights:

- It's noted the total order price are really high in states such as SP,RJ,MG and rest of the staes the total_order price is quite lower
- Although the avg_order_price of each state is quite balanced

Remarks:

- There might be **logistical or supply chain** advantages in SP, RJ, and MG that are not present elsewhere worth investigating.
- Heavy reliance on a few states could be **risky** if market dynamics shift

4.c Calculate the Total & Average value of order freight for each state

Ans

```

SELECT
    customers.customer_state,
    Round(SUM(order_items.freight_value),2) AS total_freight_value,
    Round(AVG(order_items.freight_value),2) AS average_freight_value
FROM
    `case-study-463713`.`Target`.`orders` AS orders
INNER JOIN
    `case-study-463713`.`Target`.`order_items` AS order_items
ON
    orders.order_id = order_items.order_id

```

INNER JOIN

```
`case-study-463713`.`Target`.`customers` AS customers
```

ON

```
orders.customer_id = customers.customer_id
```

GROUP BY

1

ORDER BY

```
customers.customer_state;
```

Row	customer_state	total_freight_value	average_freight_value
1	AC	3686.75	40.07
2	AL	15914.59	35.84
3	AM	5478.89	33.21
4	AP	2788.5	34.01
5	BA	100156.68	26.36
6	CE	48351.59	32.71
7	DF	50625.5	21.04
8	ES	49764.6	22.06
9	GO	53114.98	22.77
10	MA	31523.77	38.26
...	...	...	...

Insights:

- It's noted the total freight price are high in states such as SP, RJ, MG and rest of the Staes the total freight price is quite lower
- These states may serve as **major logistics hubs** or be central demand zones, naturally attracting more shipments and thus higher cumulative freight costs.

Remarks:

- States such as SP, RJ, and MG show significantly higher total freight costs, likely due to higher order volumes and urban logistics demands. In contrast, other states have noticeably lower freight totals, not necessarily due to lower costs per shipment but rather reduced shipping activity. This points to a potential opportunity to increase reach and demand in underserved states while evaluating whether **logistics in** high-cost areas can be optimized for efficiency

5.a Find the no. of days taken to deliver each order from the order's purchase date as delivery time.

Also, calculate the difference (in days) between the estimated & actual delivery date of an order.

Do this in a single query.

You can calculate the delivery time and the difference between the estimated & actual delivery date using the given formula:

- a. **time_to_deliver** = order_delivered_customer_date - order_purchase_timestamp
- b. **diff_estimated_delivery** = order_delivered_customer_date - order_estimated_delivery_date

Ans

```
SELECT
    orders.order_id,
    DATE_DIFF( orders.order_delivered_customer_date, orders.order_purchase_timestamp,
        DAY) AS time_to_deliver,
    DATE_DIFF( orders.order_delivered_customer_date,
        orders.order_estimated_delivery_date, DAY) AS diff_estimated_delivery
FROM
    `case-study-463713`.`Target`.`orders` AS orders
    where DATE_DIFF( orders.order_delivered_customer_date,
        orders.order_purchase_timestamp, DAY) is not null
    or DATE_DIFF( orders.order_delivered_customer_date,
        orders.order_estimated_delivery_date, DAY) is not null
ORDER BY
    orders.order_purchase_timestamp;
```

Row	order_id	time_to_deliver	diff_estimated_delivery
1	bfb0f9bdef84302105ad712db...	54	36
2	3b697a20d9e427646d9256791...	23	0
3	be5bc2f0da14d8071e2d45451...	24	-10
4	65d1e226dfaeb8cdc42f665422...	35	-16
5	a41c8759f7be7aab36ea07e038b...	30	-25
6	d207cc272675637bfed0062edf...	27	-22
7	cd3b8574c82b42fc8129f6d502...	10	-39
8	ae8a60e4b03c5a4ba9ca0672c...	30	-27
9	ef1b29b591d31d57c0d733746...	28	-23
10	0a0837a5eee9e7a9ce2b1fa831...	18	-32

Insights:

- Here difference between estimated and actual date is negative which shows most of the delivery made reaches late to customers

Remarks:

- Here the Target needs to improve their delivery of item to customers as this could later impact the business and the customer feeling satisfied with the Services of the brand

5b. Find out the top 5 states with the highest & lowest average freight value.

Ans

```
with stateFreight as(
    select seller_state,round(avg(freight_value),2) as average_freight_value from
    `Target.order_items` as o
    inner join `Target.sellers` as s on o.seller_id=s.seller_id
    group by seller_state order by average_freight_value
),
top_5_state as (
    select * from stateFreight order by average_freight_value desc
    limit 5
),
low_5_state as (
    select * from stateFreight order by average_freight_value
    limit 5
)
select * from top_5_state union all select * from low_5_state
```

Row	seller_state ▼	average_freight_value ▼
1	RO	50.91
2	CE	46.38
3	PB	39.19
4	PI	36.94
5	AC	32.84
6	SP	18.45
7	PA	19.39
8	RJ	19.47
9	DF	20.57
10	PR	22.72

Insights:

- The First 5 rows display the highest average freight values states, and last 5 rows displays the lowest average freight values states

Remarks:

- It is noted the freight value for the stats RO, CE, PB, PI and AC are high and these states shipment made need to be optimized for better profits

5c. Find out the top 5 states with the highest & lowest average delivery time.

Ans

```
WITH
order_delivery_time AS (
SELECT
customer_state,
round(avg(DATE_DIFF(order_delivered_customer_date, order_purchase_timestamp,
DAY)),2) AS delivery_time
FROM
`case-study-463713`.`Target`.`orders` as o
inner join `Target.customers` as c on o.customer_id=c.customer_id
WHERE
order_delivered_customer_date IS NOT NULL
AND order_purchase_timestamp IS NOT NULL
group by customer_state),
```

```

top_5_delivery as (
  select * from order_delivery_time order by order_delivery_time.delivery_time desc
  limit 5
),
low_5_delivery as (
  select * from order_delivery_time order by order_delivery_time.delivery_time
  limit 5
)
select * from top_5_delivery union all select * from low_5_delivery

```

Row	customer_state ▼	delivery_time ▼
1	RR	28.98
2	AP	26.73
3	AM	25.99
4	AL	24.04
5	PA	23.32
6	SP	8.3
7	PR	11.53
8	MG	11.54
9	DF	12.51
10	SC	14.48

Insights:

- The first 5 states (rows 1-5) display the highest average delivery time, and the last 5 state (rows 6-10) displays the lowest average delivery time

Remarks:

- Here the first 5 states such as RR, AP, AM, AL, PA takes a lot of time to delivery their items to customers thus need to improve delivery management in these by introducing more delivery officials, introducing more stores in these States etc. which could reduce average delivery time in these states.

5d. Find out the top 5 states where the order delivery is fast as compared to the estimated date of delivery.You can use the difference between the

averages of actual & estimated delivery date to figure out how fast the delivery was for each state.

Ans

```
SELECT
    customer_state,
    round(AVG( TIMESTAMP_DIFF( order_delivered_customer_date,
order_estimated_delivery_date, DAY)),2) AS avg_delivery_speed
FROM
    `case-study-463713`.`Target`.`orders` AS o
INNER JOIN
    `case-study-463713`.`Target`.`customers` AS c
ON
    o.customer_id = c.customer_id
GROUP BY
    1
ORDER BY
    avg_delivery_speed
LIMIT
    5;
```

Row	customer_state	avg_delivery_speed
1	AC	-19.76
2	RO	-19.13
3	AP	-18.73
4	AM	-18.61
5	RR	-16.41

Insights:

- These are the 5 States which makes delivery fast by an average of more than 16 days (about 2 and a half weeks) prior to the estimated delivery time

Remarks:

- Possible Overestimation in Estimated Delivery Time
- This indicates strong coordination between sellers, carriers, and local distribution centers.

6.a Find the month on month no. of orders placed using different payment types

Ans

```
SELECT
```

```

    EXTRACT(MONTH FROM orders.order_purchase_timestamp) AS
month, payments.payment_type, COUNT(*) AS num_orders
FROM
    `case-study-463713`.`Target`.`orders` AS orders
LEFT JOIN
    `case-study-463713`.`Target`.`payments` AS payments
ON
    orders.order_id = payments.order_id
GROUP BY
    month,
    payments.payment_type
ORDER BY
    month,
    payments.payment_type;

```

Row	month	payment_type	num_orders
1	1	UPI	1715
2	1	credit_card	6103
3	1	debit_card	118
4	1	voucher	477
5	2	UPI	1723
6	2	credit_card	6609
7	2	debit_card	82
8	2	voucher	424
9	3	UPI	1942
10	3	credit_card	7707
11	3	debit_card	109

Insights:

- After reviewing it can be noticed that most of the people complete their transaction using credit cards followed by UPI payments and then debit cards

Remarks:

- can implement Targets very own payment method by attracting users by giving them some credit points which can be later used to purchase other products, and avail offers and so on

6b. Find the no. of orders placed based on the payment installments that have been paid

Ans

```
SELECT
    payments.payment_installments,
    COUNT(*) as No_order_placed
FROM
    `case-study-463713`.`Target`.`payments` AS payments
LEFT OUTER JOIN
    `case-study-463713`.`Target`.`orders` AS orders
ON
    payments.order_id = orders.order_id
GROUP BY
    payments.payment_installments;
```

Row	payment_installm...	No_order_placed
1	0	2
2	1	52546
3	2	12413
4	3	10461
5	4	7098
6	5	5239
7	6	3920
8	7	1626
9	8	4268
10	9	644

Insights:

- Many people prefer to buy items through credit cards in terms of installments

Remarks:

- Give more options on making purchase on products in terms on installments